

# Formelsamling til Elektroteknik

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## Resumé

En formelsamling med formler indenfor faget elektroteknik. *Indholdsfortegnelse på næste side.*

Formelsamlingen er opdelt i 4 dele:

**Love og Regler** En liste over forskellige love som kan bruges til at udregne forskellige værdier.

**Matematik** Forskellig matematik til brug af løsning af differentiale ligninger f.eks.

**Komponenter** En samling af komponenter og informationer omkring dem.

**Kogebøger** Stens Kogebøger skrevet direkte af.

# Indhold

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# 1 Love og Regler

**Kirchoffs strømlov (KCL)**

$$I_1 + I_2 + I_3 = 0$$

**Kirchoffs spændings lov (KVL)**

$$V_1 + V_2 + V_3 = 0$$

**Effekt for passiv komponent (+  $\rightarrow$  -)**

$$P = VI$$

**Effekt for aktiv komponent (+  $\leftarrow$  -)**

$$P = -VI$$

**Ohms lov**

$$V_R = RI_R$$

## 1.1 Serieopstillinger

**Alle Strømme er ens**

$$I_t = I_1 = I_2$$

**Spændinger adderes**

$$V_t = V_1 + V_2$$

**Ækvivalent modstand**

$$R_{eq} = R_1 + R_2$$

**Spændingsdeler**

$$V_{R1} = V_t \cdot \frac{R_1}{R_1 + R_2}$$

$$V_{R2} = V_t \cdot \frac{R_2}{R_1 + R_2}$$

**Specielt, hvis  $R_1 = R_2$  da gælder**

$$V_{R1} = V_{R2} = \frac{V_t}{2}$$

## 1.2 Parallelopstilling

**Alle Spændinger er ens**

$$V_t = V_1 = V_2$$

**Strømme adderes**

$$I_t = I_1 + I_2$$

**Ækvivalent modstand**

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

### 1.2.1 Specielt for *netop to* modstande

**Ækvivalent modstand**

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

**Specielt hvis  $R_1 = R_2$  da gælder**

$$R_{eq} = \frac{R_1}{2} = \frac{R_2}{2}$$

**3 modstande (?)**

$$R_{eq} = \frac{R_1}{3} = \frac{R_2}{3} = \frac{R_3}{3}$$

**Strømdeling**

$$I_{R1} = I_t \cdot \frac{R_2}{R_1 + R_2}$$

$$I_{R2} = I_t \cdot \frac{R_1}{R_1 + R_2}$$

**Specielt hvis  $R_1 = R_2$  da gælder**

$$I_{R1} = I_{R2} = \frac{I_t}{2}$$

## 1.3 Thevenin og Norton

**Thevenin spænding er lig open circuit**

$$V_t = V_{oc}$$

**Norton strømmen er lig short circuit**

$$I_n = I_{sc}$$

**Thevenin og Norton modstand**

$$R_t = R_n = \frac{V_{oc}}{I_{sc}} = \frac{V_t}{I_n}$$

**Hvis ingen styrede kilder**

$$R_n = R_t = R_{eq} \quad \swarrow$$

**Effekt på load**

$$P_L = V_t^2 \frac{R_L}{(R_t + R_L)^2}$$

**Maks effekt**

$$P_{max} = \frac{V_t I_n}{4}$$

$$P_{max} = \frac{V_t^2}{4R_t}$$

### 1.3.1 Kildetransformation

$$R_n = R_t$$

**Strøm**  $\rightarrow$  **spænding**

$$V_t = I_n \cdot R_n$$

**Spænding**  $\rightarrow$  **strøm**

$$I_n = \frac{V_t}{R_t}$$

## 1.4 Boolsk Algebra

**1.**

$$A + B = B + A \quad ; \quad AB = BA$$

**2.**

$$A + (B + C) = (A + B) + C \quad ; \quad A(BC) = (AB)C$$

**3.**

$$A(B + C) = AB + AC \quad ; \quad (A + B)(C + D) = AC + AD + BC + BD$$

$$1. A \cdot 0 = 0$$

$$2. A \cdot 1 = A$$

$$3. A + 0 = A$$

$$4. A + 1 = 1$$

$$5. A \cdot A = A$$

$$6. A + A = A$$

$$7. A \cdot \overline{A} = 0$$

$$8. A + \overline{A} = 1$$

$$9. \overline{\overline{A}} = A$$

$$10A. A + \overline{A} \cdot B = A + B$$

$$10B. \overline{A} + A \cdot B = \overline{A} + B$$

**Demorgan 1.**

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

**Demorgan 2.**

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

**Xor**

$$A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$$

**Xnor**

$$\overline{A \oplus B} = \overline{A} \cdot \overline{B} + A \cdot B$$

## 1.5 Digitale byggeblokke

**Talrum**

$$[0; 2^n - 1]$$

**Talrum (2's complement)**

$$[-2^{n-1}; 2^{n-1} - 1]$$

## 1.6 Ikke lineære komponenter

### 1.6.1 Dioder

**Strøm som funktion af spænding for diode**

$$I_d = \frac{V_S - V_d}{R_d}$$

### 1.6.2 Transistorer

**Strøm på basis**

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

**Strøm på kollektor**

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

## 2 Matematik

$$f(x) = e^{ax} \Rightarrow \begin{cases} f'(x) = a \cdot e^{ax} \\ \int f(x) dx = \frac{1}{a} \cdot e^{ax} \\ f(0) = 1 \\ f(x) \rightarrow 0 \text{ for } x \rightarrow -\infty, a > 0 \\ f(x) \rightarrow 0 \text{ for } x \rightarrow \infty, a < 0 \end{cases}$$

## 2.1 Panserformlen

$$y'(t) + p(t) \cdot y(t) = q(t)$$

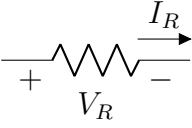
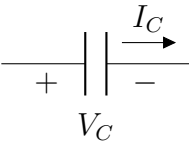
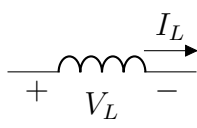
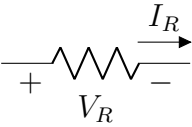
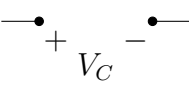
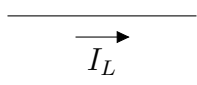
$$\Updownarrow$$

$$y(t) = e^{-P(t)} \cdot \int e^{P(t)} \cdot q(t) dt + ke^{-P(t)}$$

hvor  $k$  er en konstant og  $P(t)$  er en stamfunktion til  $p(t)$

## 3 Komponenter

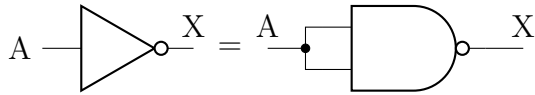
### 3.1 Lineære Komponenter, Leksikon

| Type                                                                               | Modstand ( $R$ )                                                                                      | Kondensator ( $C$ )                                                                                   | Spole ( $L$ )                                                                                         |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Symbol                                                                             |                      |                      |                    |
| Ligning                                                                            | $V_R = R \cdot I_R$                                                                                   | $I_C = C \cdot \frac{dV_C}{dt}$                                                                       | $V_L = L \cdot \frac{dI_L}{dt}$                                                                       |
| Enhed                                                                              | $[R] = \frac{V}{A} = \Omega$                                                                          | $[C] = \frac{C}{V} = \frac{AS}{V} = s \cdot \Omega^{-1} = F$                                          | $[L] = \frac{Vs}{A} = s\Omega = H$                                                                    |
| Serie                                                                              | $R_{eq} = R_1 + R_2$                                                                                  | $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \Leftrightarrow C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$ | $L_{eq} = L_1 + L_2$                                                                                  |
| Parallel                                                                           | $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Leftrightarrow R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$ | $C_{eq} = C_1 + C_2$                                                                                  | $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} \Leftrightarrow L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$ |
| $P_{øjeblik}$                                                                      | $P_\phi = V_R \cdot I_R = \frac{V_R^2}{R} = R \cdot I_R^2$                                            | $P_\phi = V_C \cdot I_C = CV_C \frac{dV_C}{dt}$                                                       | $P_\phi = V_L \cdot I_L = L \cdot I_L \cdot \frac{dI_L}{dt}$                                          |
| $P_{middel}$<br>(Sinus med amplitude)<br>$V_A$ og $V_{RMS} = \frac{V_A}{\sqrt{2}}$ | $P_m = V_{RAS} \cdot I_{RMS}$<br>$= \frac{V_{RMS}^2}{R} = \frac{V_A^2}{2R}$                           | $P_m = 0$                                                                                             | $P_m = 0$                                                                                             |
| Oplagret energi                                                                    | $\omega = 0$                                                                                          | $\omega = \frac{1}{2} C \cdot V_C^2$                                                                  | $\omega = \frac{1}{2} L \cdot I_L^2$                                                                  |
| Steady state model =<br>"efter lang tid"                                           |                    |                    |                  |
| Kontinuitet                                                                        | Ingen krav                                                                                            | $V_C(t_-) = V_C(t_+)$<br>$V_C$ er kontinuert                                                          | $I_L(t_-) = I_L(t_+)$<br>$I_L$ er kontinuert                                                          |

### 3.2 Andre gates lavet af NAND

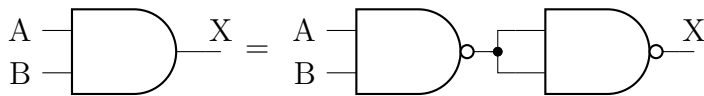
**Not**

$$X = \overline{A \cdot A} = \overline{A}$$



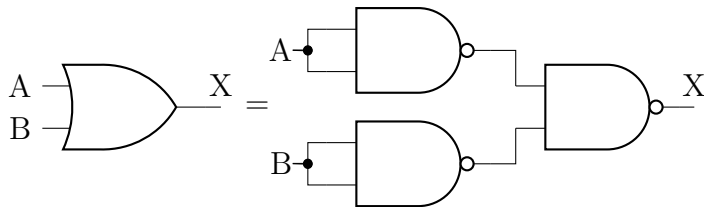
**And**

$$X = \overline{\overline{A \cdot B}} = A \cdot B$$



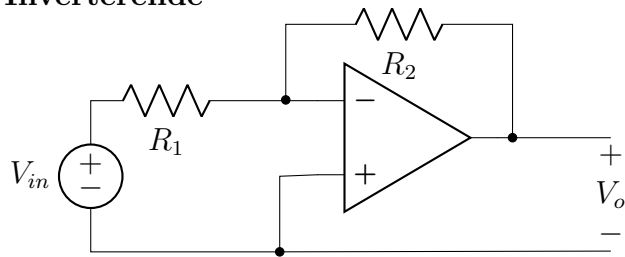
**Or**

$$X = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$



### 3.3 Opamp koblinger

**Inverterende**



$$V_o = -V_{in} \left( \frac{R_2}{R_1} \right)$$

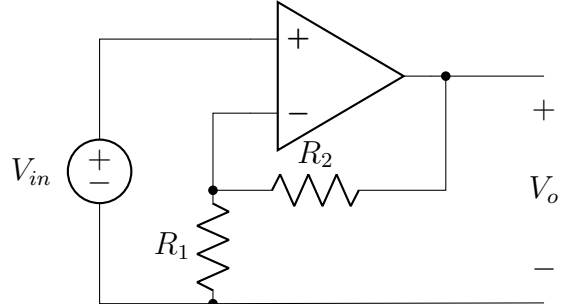
$$A_V = -\frac{R_2}{R_1}$$

$$R_{in} = \frac{V_{in}}{I_{in}} = R_1$$

$$R_o = \frac{V_{o,oc}}{I_{o,sc}} = 0$$



### Ikke Inverterende



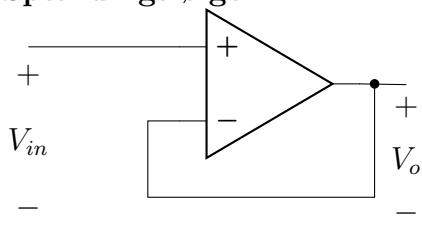
$$V_o = V_{in} \left( 1 + \frac{R_2}{R_1} \right)$$

$$A_V = 1 + \frac{R_2}{R_1}$$

$$R_{in} = \frac{V_{in}}{I_{in}} = \infty$$

$$R_o = \frac{V_{o,oc}}{I_{o,sc}} = 0$$

### Spændingsfølger



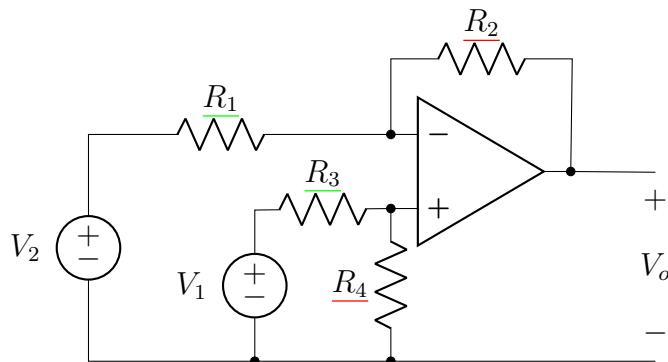
$$V_o = V_{in}$$

$$A_V = 1$$

$$R_{in} = \infty$$

$$R_o = 0$$

## Differens



$$V_o = \left( \frac{R_4(R_1+R_2)}{R_1(R_3+R_4)} \right) \cdot V_1 - \left( \frac{R_2}{R_1} \right) \cdot V_2$$

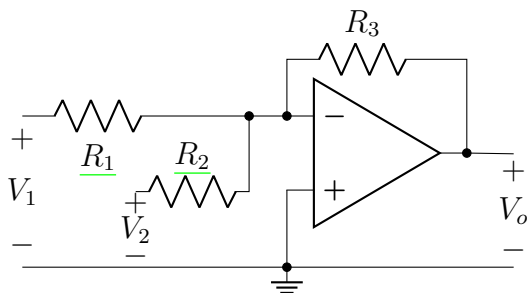
Hvis  $R_3 = R_1$  og  $R_4 = R_2$  da gælder

$$V_o = \left( \frac{R_2}{R_1} \right) (V_1 - V_2)$$

Og hvis  $R_1 = R_2 = R_3 = R_4$  da gælder

$$V_o = V_1 - V_2$$

## Inverterende sumkobling

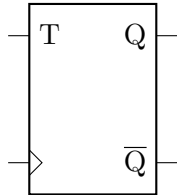


$$V_o = - \left( V_1 \frac{R_3}{R_1} + V_2 \frac{R_3}{R_2} \right)$$

Specielt, hvis  $R_1 = R_2$

## 3.4 Flip-Flops

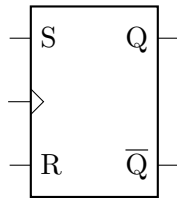
### 3.4.1 T-type



$$Q^* = T \oplus Q$$

| $T$ | $Q^*$          |
|-----|----------------|
| 0   | $Q$            |
| 1   | $\overline{Q}$ |

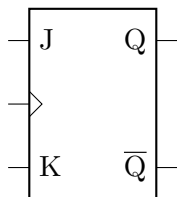
### 3.4.2 SR-type



$$Q^* = S + \overline{R}Q$$

| $SR$ | $Q^*$   |
|------|---------|
| 00   | $Q$     |
| 01   | 0       |
| 10   | 1       |
| 11   | FORBUDT |

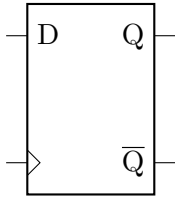
### 3.4.3 JK-type



$$Q^* = J\overline{Q} + \overline{K}Q$$

| $JK$ | $Q^*$          |
|------|----------------|
| 00   | $Q$            |
| 01   | 0              |
| 10   | 1              |
| 11   | $\overline{Q}$ |

#### 3.4.4 D-type



$$Q^* = D$$

| $D$ | $Q^*$ |
|-----|-------|
| 0   | 0     |
| 1   | 1     |

## 4 Kogebøger